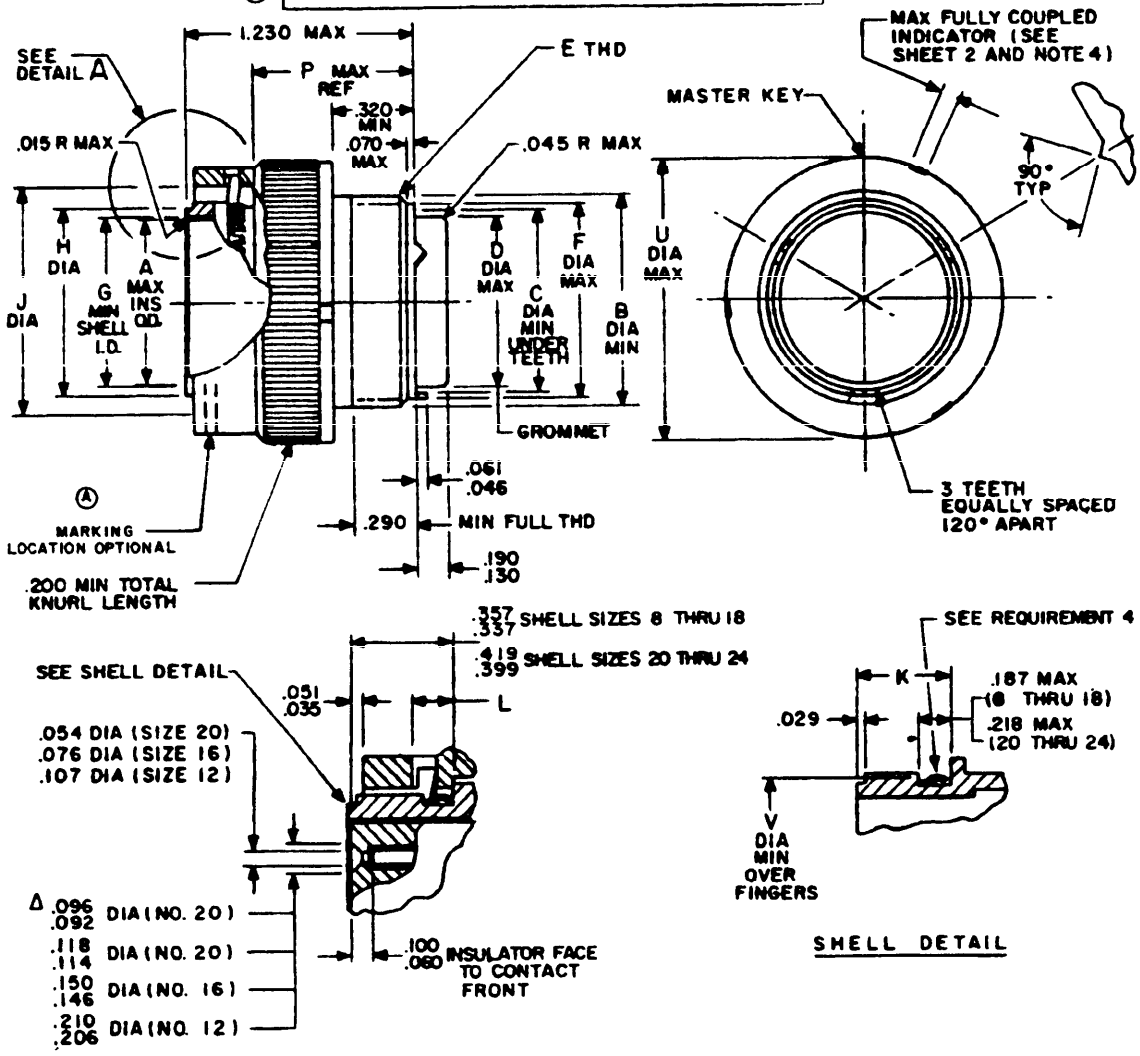
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), PLUG, (RFI GROUNDING, BAYONET COUPLING, CRIMP SOCKET CONTACT), (SERIES I, CLASSES G AND R)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Ⓐ The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-26482.

Ⓐ INACTIVE. SUPERSEDED BY MS3475 OF MIL-C-26482.



DETAIL A FIGURE 1

Δ SEE TABLE II FOR CONTACT CAVITIES REQUIRING THIS DIMENSION Ⓐ denotes changes

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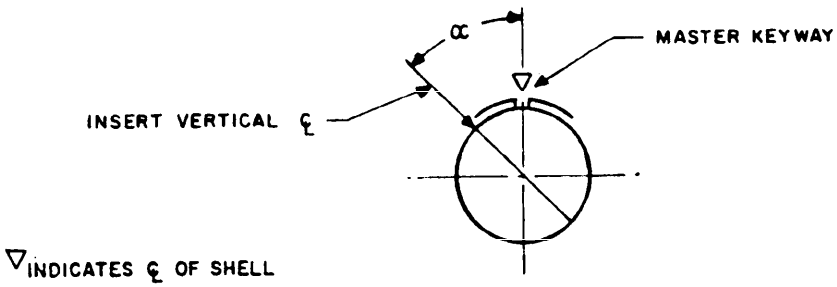
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TABLE I Connector Dimensions

SIZE	A	B	C	D	E THREAD	F	G	H	J
8	.285	.470	.370	.305	1/2-20 UNF-2A	.437	.292	.358/.352	.486/.480
10	.402	.600	.497	.405	5/8-24 UNEF-2A	.572	.409	.486/.480	.607/.601
12	.516	.724	.613	.531	3/4-20 UNEF-2A	.687	.523	.598/.592	.766/.760
14	.641	.849	.738	.665	7/8-20 UNEF-2A	.812	.648	.723/.717	.890/.884
16	.766	.974	.863	.790	1-20 UNEF-2A	.937	.772	.818/.842	1.015/1.009
18	.855	1.030	.919	.869	$1\frac{1}{16}$-18 UNEF-2A	.992	.862	.948/.942	1.141/1.135
20	.980	1.154	1.044	.994	$1\frac{3}{16}$-18 UNEF-2A	1.117	.987	1.073/1.067	1.265/1.259
22	1.105	1.279	1.169	1.119	$1\frac{5}{16}$-18 UNEF-2A	1.242	1.111	1.198/1.192	1.390/1.384
24	1.229	1.404	1.294	1.244	$1\frac{7}{16}$-18 UNEF-2A	1.367	1.237	1.323/1.317	1.515/1.509

SIZE	K	L	M	N	P	U	V	W
8	.363/.343	.133/.093	.403/.392	.581/.570	.887	.782	.369	2.334
10	↑	↑	.531/.520	.702/.691	↑	.926	.497	↑
12			.680/.669	.876/.865		1.043	.614	
14			.805/.794	1.000/.989		1.183	.739	
16	↓		.930/.919	1.125/1.114	↓	1.305	.864	
18	.363/.343		1.030/1.019	1.250/1.239	.887	1.391	.969	
20	.425/.405	↓	1.155/1.144	1.375/1.364	.825	1.531	1.094	
22	.425/.405	.133/.093	1.280/1.269	1.500/1.489	.825	1.656	1.219	↓
24	.425/.405	.172/.132	1.405/1.394	1.629/1.618	.825	1.777	1.344	2.334

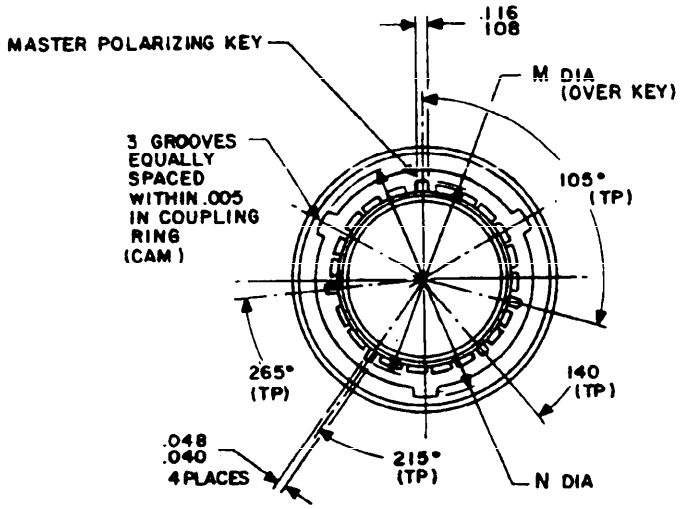
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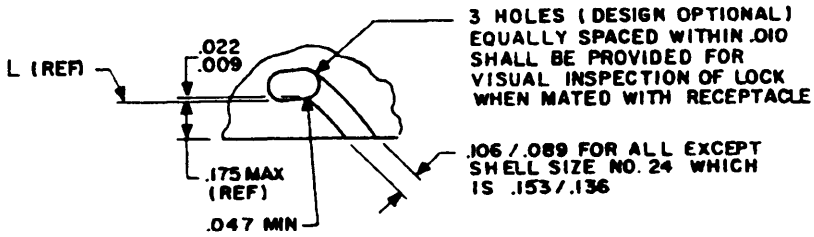


INSERT POSITION

(AS VIEWED FROM ENGAGING END)

FIGURE 2

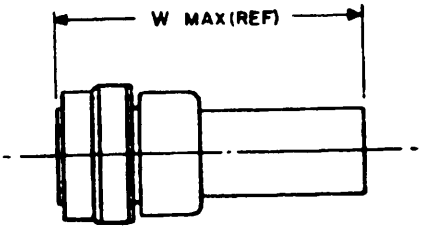




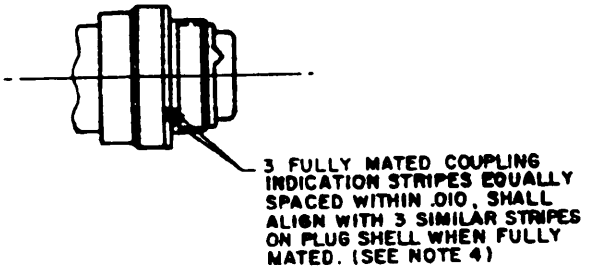
CAM GROOVE DETAIL

FIGURE 3

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ASSEMBLED TO R.F.I. BACKSHELL



COUPLED POSITION

FIGURE 4

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TABLE II. Insert arrangement requiring reduced diameters for raised seal barriers on outer row of contact cavities as indicated.

Shell	Insert arrangements	Contact cavities
8	33 and 98	A, B, and C
12	10	C and G
14	12	A,B,C,D,E,F,G, and H
14	18	A,C,E,G,J, and L
14	19	B,D,F,H,K, and M
16	26	A,B,C,D,E,F,G,H,J, K,L,M,N,P, and R
18	32	A,B,C,D,E,F,G,H,J, K,L,M,N,P,R,S, and T
22	41	A,B,C,D,E,F,G,H,J, K,L,M,N,P,R,S,T,U, V,W,X, and Y

NOTES:

1. For alternate polarizing positions, the connector insert is rotated in respect to the shell.
2. In "Normal Position" (Position "N") the insert vertical center line coincides with the center line of the master key of the shell.
3. In "Alternate positions" (Positions "W", "X", "Y", and "Z") the socket insert is rotated α degrees counterclockwise relative to the center line of the master key of the shell.
4. Fully mated coupling indication stripes and polarization stripes shall be color coded blue. Stripes shall have min. width of .050 on the coupling ring and min. width of .040 on the shells.
5. Four keys (MMC) and insert shall be located within .0035 either side of (TP) relative to master key (MMC) and shell O.D. (MMC).
6. Dimensions in inches. Unless otherwise specified, tolerance shall be .XX $\pm$ .010 and .XXX $\pm$ .005, Angular, X^{o} $\pm$ 1° -0' and X^{o} X' $\pm$ 30'.

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REQUIREMENTS:

Dimensions and configuration: See figures.

This connector mates with MIL-C-83723/2, /4, /6, /8 through /12 (USAF).

(A) For insert arrangements see MIL-STD-1669.

Rear harware must be used per MIL-C-83723/47(USAF).

RFI spring strip shall have a minimum of six fingers per inch.

Connector assembly must include a backshell to meet the moisture sealing requirements.

Material: Shell shall be aluminum alloy per QQ-A-367, coupling ring shall be aluminum alloy.

Wire: See table III.

(A) Connector identification marking shall be in accordance with MIL-C-26482. Example: MS3475L----, M83723/43R----.

TABLE III. Wire data.

CONTACT SIZE	WIRE SIZE AWG	INSULATION O.D. IN INCHES / WIRE SPECIFICATION 3/			
		MIL-W-22759/4	MIL-W-16878/4	MIL-W-81381/-	
				LIGHT WT /1	MEDIUM WT /3
20	24	-	.040/.048 1/	.037/.043	.047/.054
	22	.071/.077	.046/.054	.043/.050	.053/.060
	20	.079/.085 2/	.054/.062	.051/.058	.061/.068
16	20	.079/.085	.054/.062 1/	.051/.058	.061/.068
	18	.092/.098	.064/.074	.060/.067	.071/.079
	16	.099/.107 2/	.073/.087	.066/.074	.079/.087
12	14	.112/.120 1/	.087/.101	.080/.089	.091/.099
	12	.129/.137 2/	.106/.120	.098/.108	.107/.117

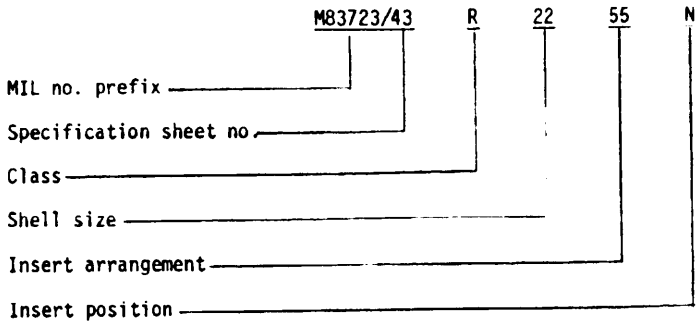
1/ WIRE TO BE USED FOR CONNECTOR QUALIFICATION.

2/ WIRE TO BE USED FOR CONTACT QUALIFICATION.

3/ LISTED WIRE SPECIFICATIONS ARE NOT ALL INCLUSIVE. WIRE WITH SMOOTH EXTERIOR SURFACE, WHICH FALLS WITHIN THE SPECIFIED RANGE AND MEETS THE DESIGNERS CRITERIA, MAY BE USED IN APPLICATION.

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Part number example:



Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17, 70, 71, 80, 82, 84
DSA - ES

Preparing activity:
Air Force - 85

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